

QKS-v2 vs ViT4Mal [2] — Full Application Latency (same scope) on IMG_DS, PYNQ-Z2

Configuration	Acc (%)	F1	Full Application Latency (ms)	FPGA Kernel Latency (us)	LUT	DSP	BRAM	Speedup vs ViT4Mal 810 ms
Q16-QKSv2-S0 (dense)	93.64	0.9621	18.5	1552	13,322	147	15	43.7x
QKSv2-S10 (12.5% pruned)	93.79	0.9631	18.8	1533	13,344	147	15	43.0x
QKSv2-S20 (18.75% pruned)	94.11	0.9651	18.6	1502	13,349	148	15	43.7x
QKSv2-S30 (31.25% pruned)	93.84	0.9634	18.7	1442	13,347	148	15	43.3x
ViT4Mal 1e2h (fastest)	810.00	—	92	—	16,545	69	92	8.8x
ViT4Mal 1e4h	870.00	—	92	—	16,029	69	84	8.8x
ViT4Mal 2e2h	1450.00	—	94	—	16,219	69	92	8.6x
ViT4Mal 2e4h (best accuracy)	1480.00	—	94	—	16,079	69	84	8.6x